

# Research Memo 1: Distribution of Communication Apprehension

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To: Project Collaborators

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Re: What is the overall distribution of communication apprehension among participants?

## Opening [↗](#)

This memo examines the overall distribution of communication apprehension among participants. Communication apprehension was measured using twelve survey items (Q1–Q6 and Q13–Q18). Reverse-coded items were recoded before calculating a composite communication apprehension score for each participant. Descriptive statistics and a histogram were used to summarize the distribution of the resulting scores.

## Descriptive Statistics

Table [Table 1](#) presents the descriptive statistics for the communication apprehension scores.

Table 1: Descriptive statistics for communication apprehension scores.

```
summary = (  
    df["communication_apprehension_score"]  
    .describe()  
    .round(2)  
    .to_frame(name="Value")  
)  
summary
```

|       | Value  |
|-------|--------|
| count | 250.00 |
| mean  | 2.41   |
| std   | 0.95   |
| min   | 1.00   |
| 25%   | 1.75   |
| 50%   | 2.09   |
| 75%   | 3.00   |
| max   | 5.00   |

## Distribution of Communication Apprehension

Figure [Figure 1](#) displays the distribution of communication apprehension scores across participants.

```
plt.figure(figsize=(6,4))  
plt.hist(  
    df["communication_apprehension_score"].dropna(),  
    bins=10,  
    edgecolor="black"
```

```

)
plt.axvline(
    df["communication_apprehension_score"].mean(),
    color="red",
    linestyle="--",
    linewidth=2,
    label="Mean"
)
plt.axvline(
    df["communication_apprehension_score"].median(),
    color="blue",
    linestyle=":",
    linewidth=2,
    label="Median"
)
)
plt.xlabel("Communication Apprehension Score")
plt.ylabel("Frequency")
plt.legend()
plt.tight_layout()
plt.show()

```

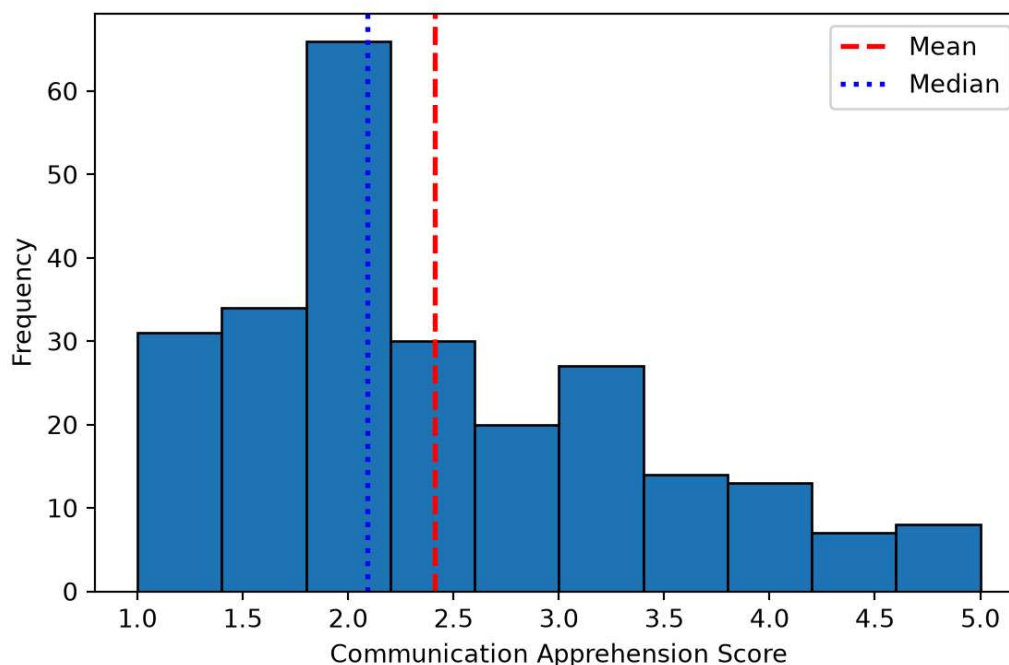


Figure 1: Distribution of communication apprehension scores.

## The Answer

As shown in [Table 1](#), the communication apprehension scores were calculated for **250 participants**. Scores ranged from **1.00** to **5.00**, with a mean of **2.41** (SD = **0.95**). The median score was **2.09**, and the middle 50% of observations fell between **1.75** and **3.00**.

Figure [Figure 1](#) shows that communication apprehension scores are concentrated toward the lower end of the scale, with relatively fewer participants reporting high levels of communication apprehension. Overall, the distribution suggests that most participants experienced **low to moderate communication apprehension**, although there is substantial variability across individuals.